PROBLEM 17 | Problems T2 defects.pdf | Crystal Defects & Solid Solutions

Cationic Vacancies in MgO from the Dissolution of Al_2O_3 and TiO_2

Original statement · Problems T2 defects.pdf

3. Calculate the number of Mg^{2+} *vacancies produced by the dissolution of: a)* 1 mole *of* Al_2O_3 *in* 99 moles *of* MgO. *(Solution:* 0.01 mol vacancies/mol MgO*) b)* 2 moles *of* TiO_2 *in* 98 moles *of* MgO. *(Solution:* 0.02 mol vacancies/mol MgO*)*

Solution: 0.01 mol vacancies/mol MgO

PHASE 1: HYPOTHESIS & PARAMETERS

Chemical Context:

The formation of substitutional solid solutions in the ionic ceramic lattice of magnesium oxide (MgO, NaCl-type structure with a cation sublattice of Mg^{2+} and an anion sublattice of O^{2-}) is studied.

Fundamental Requirement:

- **Principle of Macroscopic Electroneutrality:** The total positive charge of the incorporated cations must exactly equal the charge of the substituted cations. The valence difference between the dopant cations (Al^{3+} or Ti^{4+}) and the matrix cation (Mg^{2+}) forces the generation of **cationic vacancies** V''_{Mg} [Session 4 Slide 22].

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing slide 22 [Session 4 Slide 22]:

$$\text{Generation of defects due to solid solution in Ionic solids: } 3\,\mathrm{Mg}^{2+} \longleftrightarrow 2\,\mathrm{Al}^{3+} + 1\,V_{\mathrm{Mg}}$$

Electric Charge Balance:

1. **For $\mathrm{Al_2O_3}$:**
2. Each formula unit of $\mathrm{Al_2O_3}$ supplies $2\ \mathrm{Al}^{3+}$ cations (charge $+6$).
3. To house these 2 cations in the Mg^{2+} sublattice while maintaining neutrality, $3\ \mathrm{Mg}^{2+}$ cations (charge $+6$) must be replaced.
4. Since $2\ \mathrm{Al}^{3+}$ cations occupy 2 sites of the cation lattice, the third site is left empty as **one cationic vacancy**:

$$\text{For each mole of } \mathrm{Al_2O_3} \implies 1 \text{ mole of vacancies } V_{\mathrm{Mg}}$$

5. **For $\mathrm{TiO_2}$:**
6. Each formula unit of $\mathrm{TiO_2}$ supplies $1\ \mathrm{Ti}^{4+}$ cation (charge $+4$).
7. To balance the charge $+4$, $2\ \mathrm{Mg}^{2+}$ cations (charge $+4$) must be replaced.
8. Since the Ti^{4+} cation occupies 1 cation site, the second site remains as **one cationic vacancy**:

$$\mathrm{Ti}^{4+} + 2\,\mathrm{Mg}^{2+} \implies \mathrm{Ti}_{\mathrm{Mg}}^{\bullet\bullet} + V_{\mathrm{Mg}}'' \implies \text{For each mole of } \mathrm{TiO_2} \implies 1 \text{ mole of vacancies } V_{\mathrm{Mg}}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Dissolution of $1\ \mathrm{mol}$ of $\mathrm{Al_2O_3}$ in 99 moles of MgO

- Moles of cationic vacancies generated:

$$n_{\text{vacancies}} = 1 \text{ mol of } \mathrm{Al_2O_3} \times \left(\frac{1 \text{ mol } V_{\mathrm{Mg}}}{1 \text{ mol } \mathrm{Al_2O_3}} \right) = 1.0 \text{ mol of vacancies}$$

- Amount of substance of the matrix: $n_{\mathrm{MgO}} = 99$ moles.
- Relative concentration per mole of MgO:

$$\frac{n_{\text{vacancies}}}{n_{\text{MgO}}} = \frac{1.0 \text{ mol vacancies}}{99 \text{ moles MgO}} = 0.010101 \approx \mathbf{0.01} \text{ mol vacancies / mol MgO}$$

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2. Part b: Dissolution of 2 moles of TiO_2 in 98 moles of MgO

- Moles of cationic vacancies generated:

$$n_{\text{vacancies}} = 2 \text{ moles of } \text{TiO}_2 \times \left(\frac{1 \text{ mol } V_{\text{Mg}}}{1 \text{ mol } \text{TiO}_2} \right) = 2.0 \text{ moles of vacancies}$$

- Amount of substance of the matrix: $n_{\text{MgO}} = 98$ moles.
- Relative concentration per mole of MgO:

$$\frac{n_{\text{vacancies}}}{n_{\text{MgO}}} = \frac{2.0 \text{ mol vacancies}}{98 \text{ moles MgO}} = 0.020408 \approx \mathbf{0.02} \text{ mol vacancies / mol MgO}$$

`PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION`

- **Extrinsic vs Thermal Defects:** Unlike intrinsic thermal vacancies (which depend exponentially on temperature), these vacancies are **extrinsic**: their concentration is strictly fixed by the dopant stoichiometry and remains constant even on cooling to room temperature.
- **Technological Application:** The deliberate introduction of higher-valence cations (Al^{3+}, Ti^{4+}) generates an enormous fixed population of cationic vacancies, increasing ionic conductivity and accelerating the solid-state sintering rate of refractory aerospace ceramics by orders of magnitude.